

CallSeal

Prove your trading track record. Never leak a call.

Zero-knowledge trading calls on **Midnight**, written in **Compact**

github.com/northbit-dev/callseal · northbit-dev.github.io/callseal

Apache-2.0

THE PROBLEM

Paid trading signals are sold on track records nobody can check.

Screenshots

Winners get posted, losers get deleted. A "78 % hit rate" is whatever the seller decides to show.

Public calls leak

Posting every call openly proves the record, but gives the product away to non-payers and front-runners.

Trusted platforms

"Verified" signal sites see every call. The analyst has to trust the platform, and so does the buyer.

Telegram and Discord groups, newsletters and research desks all run on the same thing: **trust in a number that can't be audited.**

THE SOLUTION

Make the record public and keep the calls private.

- The analyst **seals** each call as a commitment on Midnight.
- After the close, they **prove a hit in zero knowledge**, without revealing asset, side or level.
- Every call that is not proven in time becomes a **public MISS**. Hiding losers is impossible.
- Paying subscribers can get the opening privately and **check it against the chain**.

	PUBLIC LEDGER	ANALYST'S DEVICE
analyst	key, calls, hits, misses	secret key
call	commitment, rounds, status	asset, side, level, salt
oracle	closes per round	oracle key

What leaks by design is the track record itself: who called, when, and whether it hit.

HOW IT WORKS · ZK AND COMPACT

Four steps, each one a Compact circuit.

1

Seal

`sealCall` stores H(author, H(opening)). The proof also checks the level is at least 2 % from the last close, so trivial calls can't farm hits.

2

Publish

`publishCloses`: only the oracle key can post the closes for a round. The oracle never sees any call.

3

Prove hit

`proveHit` opens the commitment inside the proof and checks it beats the expiry close. The chain learns only HIT.

4

Expire

`expireCall`: after the grace window, anyone can turn an unproven call into a MISS.

Enforced inside the proof: commitments are bound to their author, so copies can't be proven · calls can't be edited after sealing · no proving before the close or after the window · no double settlement · hedging both sides still scores a MISS. `proveHit` reads the full close vector, so its public transcript is identical whatever asset the call was on.

ARCHITECTURE

Secrets stay on the device. Only proofs reach the chain.

Analyst device

Private state: secret key and call openings. Witnesses `localSecretKey`, `pendingCall`, `storedOpening` feed the circuits.

→

Proof server

Generates the ZK proof from the compiled circuit (compactc 0.31.1, proof server 8.1.3). Only public outputs leave it.

→

Midnight ledger

Contract with 6 circuits: commitments, rounds, closes, status and per-analyst stats. Read back through the indexer.

Contract

`contract/src/callseal.compact`

Shared logic

witnesses, commitments and a simulator, used by Node and the browser

Clients

static web demo, CLI demo, network runner using midnight-js 4.1.1

ON-CHAIN PROOF · MIDNIGHT PREVIEW TESTNET

A full season, 10 transactions, all SucceedEntirely.

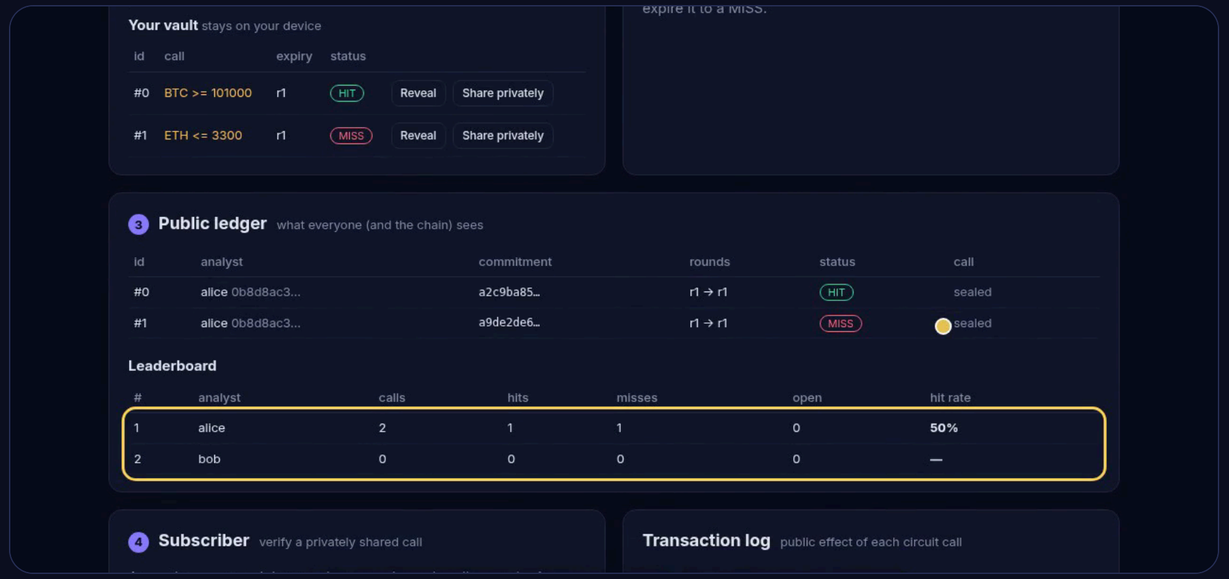
Contract `5a5f66f9c70bad0470f50716dcdf130a33f3f40932df387e9d423ac7fa4e6db0` · preview.midnightexplorer.com

STEP	TX HASH	BLOCK
deploy	`ca04ff67ce5be0aacf3704eb5d9609957816d7f343fa36b57c29aaeb5b9749d7`	1517967
oracle.publishCloses (round 0)	`0a330c0a9e3b0439eacedd26f74f4a5abdb0923cd2a42a257a0892fe6b13fba1`	1517970
alice.registerAnalyst	`59c0aebcddcee7456b8d18427255abed21fcc97b955d887fad69a829d7dfdcde`	1517973
alice.sealCall #0	`ac4b3e968914a5f4da2c5087aa4dd545dccf1a02438179d6d9829e9ba65b1f05`	1517976
alice.sealCall #1	`25cc5aedbfebb2fe60e137396b66e35069ee59db6c8e5f1839ce81c1a6096796`	1517979
oracle.publishCloses (round 1)	`9dcd644b401a6c4a8df2a40da35bd945979926bfae84c221656aae0a7d5e7dfa`	1517982
alice.proveHit #0 HIT	`74f248faa0cf23567b77f8cd97b71169a8591281a2968d8b58434376b4fe7f59`	1517985
oracle.publishCloses (round 2)	`3f177e02eefdddfc70dcd63332a9adec3e54a56e9f6b3a2c5e0349011c8809a6`	1517988
anyone.expireCall #1 MISS	`f6756f6f7534b7fef0204ff6c117ce7b394d7072deacde51dd6a33000ac6806d`	1517991
alice.revealCall #0 (optional)	`da80936ba0dcc0a6436e5725747474d8aec5dcd4e9b22e8c72ea1d5d37180004`	1517995

Every hash checked against the preview indexer · full list in docs/onchain-proof.md

DEMO

The real circuits, running in your browser.



- Seal a call; a trivial one is rejected by the circuit.
- Publish closes as the oracle.
- Prove a hit: HIT, the call stays sealed.
- Expire the loser: MISS.
- Leaderboard: 2 calls, 1 hit, 1 miss, **50 %**.
- Verify a shared opening; an edited level fails.

northbit-dev.github.io/callseal
No wallet needed. Nothing leaves the tab.

WHAT'S NEW · WHAT'S NEXT

An honest hit rate, without giving the product away.

What's new

- Selective disclosure as the product: record public, calls private
- Hidden losers still count, enforced by the protocol
- Anti-gaming rules live inside the proofs
- 27 tests, including privacy leak scans
- Full season on preview, plus a DUST fee-balancing fix

Next steps

- k-of-n oracle committee or an on-chain price feed
- Paid access to openings with shielded tokens
- Season registry and minimum call counts against key farming
- Lace wallet in the web app
- Batched windows to hide timing; more assets

Signal groups · research newsletters · prop-firm hiring on a verifiable record · copy trading